

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I **M. Meghana (192525435)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M. Meghana*

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

Unified Big Data Architecture for Retail Organization

A multinational retail company collects customer data from online websites, mobile apps, physical stores, social media and loyalty cards. A unified big-data architecture combines all this data.

Architecture:

Data Sources → Data Ingestion → Data Lake → Processing → Analytics → Real-Time Recommendations

1. Data Sources: Collect sales, customer, website and social-media data.
2. Data Ingestion: Tools such as Kafka collect real-time data.
3. Data Storage: Hadoop or cloud storage stores large amounts of structured and unstructured data.
4. Data Processing: Apache Spark processes data quickly using distributed computing.
5. Analytics & Recommendation: Machine learning analyses customer behaviour and provides personalized recommendations.

Example: If a customer searches for shoes, the system can immediately recommend matching products.

A unified architecture provides real-time personalization, better customer service and increased sales.

2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.

Distributed Computing for Scientific Research

Distributed computing divides a large task among **multiple computers or nodes** and processes them simultaneously.

1. Parallel Processing: Large scientific datasets are divided into smaller tasks and processed by multiple machines.
2. Faster Processing: Multiple computers work simultaneously, reducing execution time.
3. Scalability: More nodes can be added when the data or workload increases.

4. Fault Tolerance: Failure of one node does not necessarily stop the complete system.
5. Cost Efficiency: Multiple commodity/cloud computers can be used instead of one very powerful machine.

Example: Climate researchers can distribute millions of weather records across different computers for faster analysis.

Distributed computing provides better speed, scalability and reliability than centralized computing for large-scale scientific research.

3.Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Big Data Fraud Detection Framework

A digital payment platform can use big-data analytics to detect fraudulent transactions in real time.

Architecture:

Transactions → Kafka → Stream Processing → ML Model → Risk Score → Alert/Block

1. Data Collection: Collect transaction amount, location, device and time.
2. Real-Time Processing: Streaming tools analyze transactions immediately.
3. Feature Analysis: Compare current transactions with previous customer behavior.
4. Machine Learning: ML models calculate the probability of fraud.
5. Decision: High-risk transactions are blocked or sent for verification.

Example: A sudden high-value transaction from an unusual location can be identified as suspicious.

Real-time analytics improves fraud detection accuracy and reduces financial losses.

4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Challenges of Multimedia Big Data

Multimedia big data consists of images, videos and audio, which require large storage and processing resources.

1. **Huge Volume:** High-quality videos and images consume large storage space.
2. **High Processing Requirement:** Multimedia analysis requires high computational power.
3. **Different Formats:** Data may exist in JPEG, PNG, MP4, WAV and many other formats.
4. **High Bandwidth:** Transferring large multimedia files requires high network bandwidth.
5. **Suitable Technologies:** HDFS/cloud object storage can store data, while Spark can process it efficiently.

Example: A video streaming platform must store and process millions of videos.

Distributed storage, compression, cloud storage and parallel processing help manage multimedia big data efficiently.

5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience

Disaster Recovery and Business Continuity

Disaster recovery ensures that a cloud-based big-data platform can continue operating after failures.

Strategy:

Primary System → Replication → Backup → Disaster Recovery Site → Failover

1. **Data Replication:** Maintain copies of important data in different servers or regions.
2. **Regular Backup:** Take frequent backups and store them separately.
3. **Redundancy:** Use multiple servers and storage systems to avoid a single point of failure.
4. **Automatic Failover:** Switch automatically to a backup server when the primary server fails.

5. Monitoring and Testing: Continuously monitor the system and regularly test recovery procedures.

Example: If the primary cloud server fails, the system switches to a replicated backup server.

Disaster recovery reduces downtime and data loss and maintains business continuity.

6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Data Quality Issues in Big Data Analytics

Data quality is important because incorrect data can produce incorrect analytical results.

1. Inconsistency: The same information may be stored in different formats.
2. Duplication: Repeated records can produce incorrect results.
3. Missing Values: Missing customer or transaction information reduces analysis accuracy.
4. Data Cleaning: Remove duplicate records and correct inconsistent values.
5. Data Governance: Use validation rules, access control, metadata management and continuous monitoring.

Example: Chennai, CHENNAI and Chennai City may represent the same location and should be standardized.

Good data governance improves accuracy, reliability and decision-making.

7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behaviour analytics and distributed processing enhance recommendation accuracy and user engagement.

Scalable Recommendation System

An online streaming service can use big-data technologies to recommend movies or videos based on user behaviour.

Architecture:

User Data → Storage → Processing → Recommendation Model → Personalized Content

1. Data Collection: Collect watch history, searches, ratings and likes.
2. Data Storage: Store large-scale user data using distributed/cloud storage.
3. Behaviour Analysis: Analyse user preferences and viewing patterns.
4. Recommendation Algorithms: Use collaborative filtering, content-based filtering or hybrid methods.
5. Real-Time Updates: Continuously update recommendations when user behaviour changes.

Example: If a user frequently watches action movies, the system recommends similar action movies.

Recommendation systems improve user experience, engagement and customer retention.

8.Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Role of Cloud Computing in Big Data

Cloud computing provides on-demand storage, processing and networking resources for big-data applications.

1. Scalability: Resources can be increased or decreased according to workload.
2. Cost Efficiency: Organizations pay mainly for the resources they use.
3. Flexibility: Big-data tools can be deployed quickly.
4. High Availability: Cloud providers offer backup and redundant infrastructure.
5. Security Challenges: Privacy, unauthorized access, compliance and vendor lock-in must be managed.

Example: An e-commerce company can increase cloud resources during a festival sale and reduce them afterward.

Cloud computing makes big-data systems scalable, flexible and cost-effective

9.Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

Real-Time Analytics for Transportation and Logistics

Big-data analytics can improve transportation by analyzing GPS, traffic, vehicle and delivery data in real time.

Architecture:

GPS/Sensors → Data Ingestion → Stream Processing → Analytics → Decision → Driver

1. Real-Time Data: Collect vehicle location and traffic information using GPS and sensors.
2. Stream Processing: Process incoming data continuously.
3. Route Optimization: Identify the fastest and most efficient route.
4. Predictive Analytics: Predict traffic, delivery delays and vehicle maintenance requirements.
5. Decision Making: Send route changes and alerts to drivers and managers.

Example: If a road becomes congested, the system recommends an alternative route.

Real-time analytics reduces travel time, fuel consumption and delivery delays.

10.Critically analyse ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Ethical and Legal Challenges of Big Data

Big-data analytics provides useful information but creates privacy, security and legal concerns.

1. Privacy: Personal information should not be collected or used without proper consent.
2. Data Security: Sensitive data must be protected from unauthorized access and cyberattacks.
3. Algorithmic Bias: Biased data can result in unfair decisions.
4. Transparency: Organizations should explain how customer data is collected and used.
5. Legal Compliance: Organizations must follow applicable data-protection and privacy regulations.

Example: A company using customer location data for advertising should clearly inform users and obtain appropriate consent.

Responsible data governance ensures privacy, fairness, legal compliance and user trust.

